

CRISP DM Audit

ChatGPT

CRISP-DM Chunk: Deployment Readiness

The deployment workflow is complete and tested. It preserves Histogram GB as the official champion, requires the Extra Trees hurdle as a challenger, and prevents incompatible or multi-season files from being scored accidentally.

Complete executable checkpoint:

Key production artifacts:

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1. What was implemented

The important new deployment controls are:

```
Python
def validate_scoring_input(raw, required_predictors):
    missing = sorted(set(required_predictors).difference(raw.columns))
    if missing:
        raise ValueError(f"Production schema gate failed; missing predictors: {missing}")
    seasons = raw["season"].dropna().unique()
    if len(seasons) != 1:
        raise ValueError(f"Production season gate failed; "
                          f"expected one season, found {sorted(seasons.tolist())}")
    return int(seasons[0])
```

The models were refitted through 2022 only after the locked evaluation was completed:

```
Python
production_models = fit_locked_specifications(prepared)
package = {
    "production_models": production_models,
    "contract": contract,
    "trained_through": 2022,
    "minimum_minutes": 100,
    "required_raw_predictor_columns": raw_predictor_columns,
}
joblib.dump(package, "deployment_outputs/nba_mvp_production_models_through_2022.joblib")
```

This refit does not revise the reported 2019–2022 test results. It is the operational version that could score a future schema-compatible season.

2. Shadow-deployment result

A fixed-specification rehearsal trained through 2021 and scored 2022.

Rank	Histogram GB	Predicted share	Extra Trees hurdle	Predicted share
1	Nikola Jokić	0.6963	Nikola Jokić	0.6250
2	Giannis Antetokounmpo	0.5491	Giannis Antetokounmpo	0.4774
3	Joel Embiid	0.3564	Joel Embiid	0.3374
4	Chris Paul	0.2688	Luka Dončić	0.2091
5	Luka Dončić	0.2062	Ja Morant	0.1228

Both models placed the actual 2022 MVP, Nikola Jokić, first. Their top-five lists overlapped on four players.

This is an operational rehearsal, not new independent evidence. The 2022 outcome has already been examined during the project, and a single season should not replace the locked four-season comparison.

3. Monitoring audit

Monitor	Observed	Result
Required schema present	Yes	Pass
Candidate-count deviation	8.77%	Pass
High-PSI core features	0	Pass
Team-context median shift	1.321 recent IQRs	Pass
Champion/challenger top-1 agreement	Yes	Pass
Top-five overlap	4 players	Pass
HGB clipping	78.23%	Review

The clipping result is not a newly discovered defect. It is the previously identified tendency for HGB to produce small negative predictions for zero-share players before bounding predictions to `[0, 1]`.

4. The `win_loss_pct` monitoring decision

Raw `win_loss_pct` had PSI = 1.0369, apparently suggesting major drift. Investigation showed that this variable also produced high PSI repeatedly from 2019 through 2021.

The cause is structural:

- team winning percentages are discrete rather than smoothly distributed;
- the same value is repeated for several teammates;
- TOT rows have team context set missing;
- yearly distributions change with team records and player movement.

The 2022 seasonal median was 0.537, versus a recent-season median of 0.500. That was only 1.321 recent IQRs away and therefore passed the separate team-context check.

Final choice: retain `win_loss_pct` in the model, display its PSI, but exclude it from the automatic multi-feature PSI count. Monitor its median and missingness separately.

Benefit: this avoids systematic false alarms while retaining an important MVP predictor.

Risk: a genuine change in team-performance relationships might be detected more slowly.

Mitigation: its raw PSI, median shift, missingness change, and champion–challenger disagreement remain visible.

5. Post-outcome operational audit

These outcomes would only be calculated after MVP voting becomes available.

Model	All-player RMSE	Recipient RMSE	NDCG@5	Winner rank
Histogram GB	0.0255	0.1595	0.9396	1
Extra Trees hurdle	0.0248	0.1555	0.9450	1

Extra Trees performed slightly better in this one-season rehearsal. That does not overturn the locked model selection:

- one season is insufficient for replacement;
- HGB won the prespecified multi-season numerical comparison;
- Extra Trees was already retained specifically because of its ranking and calibration strengths.

The correct operational response is to publish both outputs, not select whichever model happened to win the latest season.

6. Interface and leakage verification

Actual self-test output:

```
Missing-column rejection    PASS Multiple-season rejection    PASS Target-free scoring    PASS - Season 2022; 496 eligible rows    Target present in features    False Serialized package size    15.97 MB
```

The production scorer can therefore work without `award_share`. The target is only needed during model fitting and retrospective outcome evaluation.

7. Retraining policy

Diagram options

My recommended policy is:

- Schema incompatibility: immediate hard stop.
- Champion/challenger disagreement: analyst review, not automatic model switching.
- One weak season: warning and investigation.
- Two consecutive materially weak seasons: consider retraining and revalidation.
- Model replacement: require a new chronological comparison against the current champion and WS baseline.

The two-season rule reduces reactions to random season-to-season variation. Its risk is slower adaptation to a genuine abrupt change, which is why schema failures and severe input anomalies remain immediate-review events.

Process map

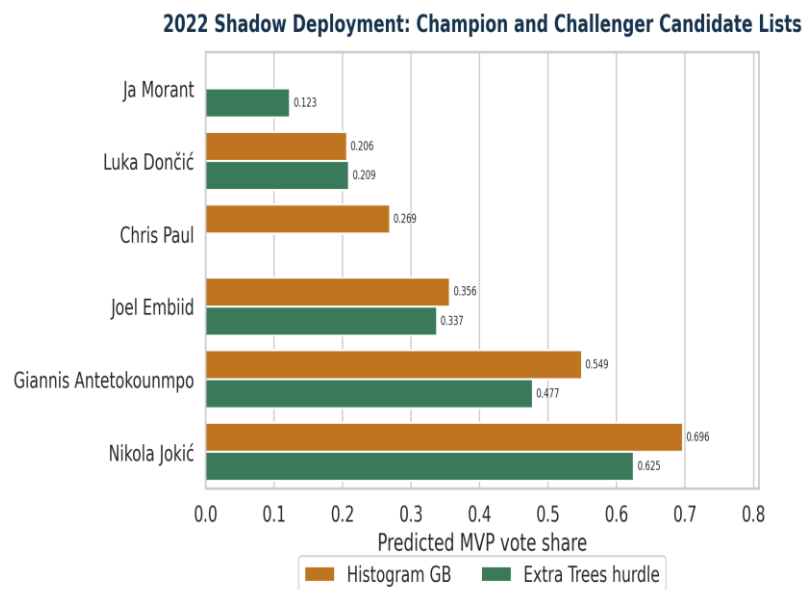
Diagram options

Completed in this chunk:

- Built the reusable champion–challenger scoring workflow.
- Fitted production models through 2022 without changing locked evidence.
- Enforced schema and one-season gates.
- Verified target-free scoring and absence of leakage.
- Investigated the misleading `win_loss_pct` PSI warning.
- Established monitoring and retraining rules.
- Produced and visually checked three professional deployment figures.

Next chunk requirements:

- Create the final audience-facing presentation.
- Emphasize the research question, major EDA findings, model comparison, final choice, risks, and recommendations.
- Keep charts readable and avoid overloading slides.
- Preserve the distinction between locked test evidence and the production refit.
- Include speaker-ready interpretations.
- Stop after the presentation checkpoint for review.



ChatGPT

No files were created or modified in this chunk. The project analysis is substantively complete, but the submission package is not yet GitHub-ready: the notebook, README, requirements file, Medium draft, YouTube outline, project transcript export, source-attribution file, and Part 2 work are missing. The local saved model artifact also fails to load and must be repaired.

Verification summary

Check	Verified result
Python source files	22
Python syntax/parse failures	0
Reproducible notebooks	0
Analytical/deployment charts	47
Report-render page images	59 intermediate/final pages
CSV result tables	5
JSON audit/contract files	4
DOCX reports	1
Rendered report PDFs	5 versions
Model artifacts	1 local file, currently unreadable
Git repository initialized	No
README.md	Missing
requirements.txt	Missing
.gitignore	Missing
Medium article draft	Missing
YouTube outline	Missing
NBA-project chat export	Missing
Part 2 replication files	Missing

The authoritative assignment PDF says Assignment 1 requires:

- Kaggle-based data science using ChatGPT.
- Exporting and submitting the chat transcript.
- Publishing and submitting a Medium report.

The expanded project brief adds the GitHub-ready notebook, source files, requirements, charts, tables, README, report, limitations/model card, Medium draft, YouTube outline, reproduction instructions, data-source attribution, and `.gitignore`.

1. GitHub repository artifacts

A. Analysis code present

All 22 Python files parse successfully, but they are separate chronological scripts rather than one consolidated pipeline or notebook.

Exact path	Purpose
/workspace/scratch/3e57ea49b558/audit_chunk1.py	Initial dataset, season, and target-distribution audit
/workspace/scratch/3e57ea49b558/audit_chunk2.py	Missingness, duplicate, repeated-name, and unit-of-observation analysis
/workspace/scratch/3e57ea49b558/audit_chunk3.py	Logical consistency and TOT-row investigation
/workspace/scratch/3e57ea49b558/eda_chunk4.py	Participation thresholds and advanced-metric stability
/workspace/scratch/3e57ea49b558/eda_chunk5.py	Candidate groups, vote-share relationships, and team success
/workspace/scratch/3e57ea49b558/eda_chunk6.py	Era trends and positional representation
/workspace/scratch/3e57ea49b558/eda_chunk7.py	Correlation, redundancy, and VIF analysis
/workspace/scratch/3e57ea49b558/eda_chunk8.py	Final Data Understanding readiness audit
/workspace/scratch/3e57ea49b558/data_prep_chunk1.py	Audited base-table preparation
/workspace/scratch/3e57ea49b558/data_prep_chunk2.py	Era-aware and season-relative features
/workspace/scratch/3e57ea49b558/data_prep_chunk3.py	Chronological design and feature-family audit
/workspace/scratch/3e57ea49b558/pre_modeling_signoff.py	Final feature, target, leakage, and preprocessing contract
/workspace/scratch/3e57ea49b558/modeling_chunk1_baselines.py	Zero, mean, and basketball-statistic baselines
/workspace/scratch/3e57ea49b558/modeling_chunk2_regularized.py	Weighted Ridge and regularized models
/workspace/scratch/3e57ea49b558/modeling_chunk3_nonlinear.py	Nonlinear ensemble models and feature importance
/workspace/scratch/3e57ea49b558/modeling_chunk4_two_stage.py	Zero-inflation and hurdle-model analysis
/workspace/scratch/3e57ea49b558/modeling_chunk5_clustering.py	Clustering analysis and predictive-increment test
/workspace/scratch/3e57ea49b558/final_comparison_checkpoint.py	Validation-stage final model comparison
	Locked 2019–2022 test evaluation
	Compatibility gate for future datasets
	Production refit, scoring gates, monitoring, and shadow deployment
/workspace/scratch/3e57ea49b558/build_final_report.py	Builds the final DOCX report

Generated Python caches, not submission artifacts:

- /workspace/scratch/3e57ea49b558/__pycache__/final_locked_test_evaluation.cpython-312.pyc
- /workspace/scratch/3e57ea49b558/__pycache__/future_data_schema_audit.cpython-312.pyc

Assessment

- Complete analytical stages: present.
- Syntax validity: verified.
- Single reproducible end-to-end program: missing.
- Jupyter notebook with narrative, code, outputs, and figures: missing.
- Full clean-environment rerun: not verified in this read-only audit.

B. Dataset

Exact path	Purpose/status
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/workspace/scratch/3e57ea49b558/upload/nba mvp stats(4).zip	Uploaded source archive; contains one file, NBA_Dataset.csv
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The public repository should probably not include this ZIP unless redistribution is clearly permitted. No Kaggle URL or attribution was found in any local Markdown, Python, JSON, or text file.

C. Saved analytical charts

There are 44 EDA/modeling charts and 3 deployment charts.

Directory	Exact filenames
/workspace/scratch/3e57ea49b558/chunk1_outputs/	target_distribution.png, positive_vote_getters_by_season.png
/workspace/scratch/3e57ea49b558/chunk2_outputs/	duplicate_and_missingness_diagnostics.png
/workspace/scratch/3e57ea49b558/chunk3_outputs/	logical_consistency_and_tot.png
/workspace/scratch/3e57ea49b558/chunk4_outputs/	advanced_metric_stability_by_minutes.png, participation_and_filter_tradeoffs.png
/workspace/scratch/3e57ea49b558/chunk5_outputs/	candidate_profile_comparison.png, mvp_share_association_ranking.png, positive_vote_share_relationships.png, team_success_and_mvp_votes.png
/workspace/scratch/3e57ea49b558/chunk6_outputs/	position_and_mvp_representation.png, raw_vs_season_percentile_associations.png, statistical_environment_by_season.png
/workspace/scratch/3e57ea49b558/chunk7_outputs/	predictor_correlation_heatmap.png, strongest_predictor_correlations.png, variance_inflation_factors.png
/workspace/scratch/3e57ea49b558/chunk8_outputs/	data_understanding_readiness_dashboard.png
/workspace/scratch/3e57ea49b558/chunk9_outputs/	audited_base_table_preparation.png
/workspace/scratch/3e57ea49b558/chunk10_outputs/	era_shift_trends.png, raw_vs_season_percentile_associations.png
/workspace/scratch/3e57ea49b558/chunk11_outputs/	chronological_evaluation_design.png, feature_family_audit.png
/workspace/scratch/3e57ea49b558/chunk12_outputs/	pre_modeling_data_contract.png
/workspace/scratch/3e57ea49b558/chunk13_outputs/	baseline_validation_comparison.png
/workspace/scratch/3e57ea49b558/chunk14_outputs/	continuous_target_weighting.png, regularized_validation_comparison.png
/workspace/scratch/3e57ea49b558/chunk15_outputs/	extra_trees_feature_importance.png, nonlinear_cv_tradeoff.png, nonlinear_validation_diagnostics.png
/workspace/scratch/3e57ea49b558/chunk16_outputs/	two_stage_validation_performance.png, two_stage_weighting_effect.png, vote_gate_diagnostics.png
/workspace/scratch/3e57ea49b558/chunk17_outputs/	cluster_count_diagnostics.png, cluster_mvp_concentration.png, cluster_predictive_increment.png, cluster_profile_heatmap.png
/workspace/scratch/3e57ea49b558/chunk18_outputs/	final_validation_pareto.png, final_validation_ranking_comparison.png, final_validation_regression_comparison.png, final_validation_winner_rank_heatmap.png
/workspace/scratch/3e57ea49b558/chunk19_outputs/	, , ,
/workspace/scratch/3e57ea49b558/deployment_outputs/	, ,

Assessment

Saved charts are extensive and satisfy the visualization requirement. A final repository should select the strongest figures rather than include every intermediate diagnostic without organization.

D. Saved result tables and metadata

Exact path	Purpose
/workspace/scratch/3e57ea49b558/chunk22_outputs/original_schema_audit.json	Original-dataset schema audit
/workspace/scratch/3e57ea49b558/deployment_outputs/deployment_contract.json	Operational model contract
/workspace/scratch/3e57ea49b558/deployment_outputs/rolling_feature_psi_history.csv	Historical feature-drift context
/workspace/scratch/3e57ea49b558/deployment_outputs/shadow_2022_feature_drift.csv	2022 drift results
/workspace/scratch/3e57ea49b558/deployment_outputs/shadow_2022_monitoring.csv	Deployment health checks
/workspace/scratch/3e57ea49b558/deployment_outputs/shadow_2022_post_outcome_metrics.csv	Shadow-run evaluation
/workspace/scratch/3e57ea49b558/deployment_outputs/shadow_2022_predictions.csv	Row-level shadow predictions
/workspace/scratch/3e57ea49b558/final_package/a11y_audit.json	Initial report accessibility audit
/workspace/scratch/3e57ea49b558/final_package/a11y_audit_final.json	Final accessibility audit; zero findings

Missing saved tables

Important tables exist in printed output or the DOCX, but not as clean repository CSV files:

- Final validation model comparison.
- Locked test model comparison.
- Season-by-season MVP ranking table.
- Predicted top-three candidates by season.
- Feature-importance table.
- Data-cleaning decision log.
- Baseline comparison table.
- Cohort-sensitivity table.

Therefore, “saved result tables” is only **partially satisfied**.

E. Model artifact

Exact path	Verification
/workspace/scratch/3e57ea49b558/deployment_outputs/nba_mvp_production_models_through_2022.joblib	Exists, but fails to load

Actual failure:

```
ValueError: EOF: reading array data, expected 34880 bytes got 33168
```

This artifact is truncated or corrupted and is not currently reproducible. It must not be presented as a working saved model.

A larger saved copy is present at the persistent path , but it was not downloaded or load-tested because this chunk was restricted to no file modifications.

F. Deployment workflow

Present:

-
- /workspace/scratch/3e57ea49b558/deployment_outputs/deployment_contract.json
- /workspace/scratch/3e57ea49b558/deployment_outputs/shadow_2022_predictions.csv
- /workspace/scratch/3e57ea49b558/deployment_outputs/shadow_2022_monitoring.csv
- /workspace/scratch/3e57ea49b558/deployment_outputs/shadow_2022_feature_drift.csv
- /workspace/scratch/3e57ea49b558/deployment_outputs/rolling_feature_psi_history.csv
- /workspace/scratch/3e57ea49b558/deployment_outputs/shadow_2022_post_outcome_metrics.csv
- Three deployment figures listed above.

Status: workflow code and monitoring outputs are present, but the corrupted joblib prevents calling the deployment package complete.

G. Report and reproduction documentation

Exact path	Status
	Valid Microsoft Word document
	Valid 12-page final rendering
	Partial reproduction handoff

The DOCX archive passed integrity testing. The final accessibility audit contains zero findings.

The handoff documents the research question, protocol, main metrics, final-evaluation command, future schema audit, and limitations. It is useful but does not replace the required README.

H. Report-rendering intermediates

These are quality-control files, not final GitHub deliverables:

- /workspace/scratch/3e57ea49b558/final_package/rendered/ One draft PDF and page-1.png through page-11.png.
- /workspace/scratch/3e57ea49b558/final_package/rendered_v2/ One draft PDF and page-1.png through page-12.png.
- /workspace/scratch/3e57ea49b558/final_package/rendered_v3/ One draft PDF and page-1.png through page-12.png.
- /workspace/scratch/3e57ea49b558/final_package/rendered_v4/ One pre-final PDF and page-1.png through page-12.png.
- /workspace/scratch/3e57ea49b558/final_package/rendered_final/ Final PDF and page-1.png through page-12.png.

Only the final DOCX and final PDF should normally enter the polished repository.

GitHub missing-items checklist

- Create an Assignment-1/ repository directory.
- Initialize a Git repository.
- Create the reproducible .ipynb notebook.
- Create one coherent run order or master pipeline.
- Create requirements.txt with pinned or bounded versions.
- Create detailed README.md.
- Create .gitignore.
- Create Kaggle source/attribution file with the exact URL.
- Add legal/download instructions instead of committing the dataset by default.
- Save the major analytical result tables as CSV.
- Repair and load-test the model artifact.
- Add a clear AI-assistance disclosure.
- Add Medium and YouTube URL placeholders.
- Organize figures into a stable reports/figures/ structure.

- Decide whether to add a standalone MODEL_CARD.md; limitations already exist in the report.
 - Verify a clean end-to-end execution from a fresh environment.
 - Push the finished repository to GitHub.
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2. Medium article work

Verified article draft: **none**.

There is no file named or functioning as:

- medium_article.md
- article_draft.md
- Medium publication draft
- article image-selection plan

The DOCX report provides source material, but it is not formatted as a Medium article.

Missing:

- Reader-friendly title and subtitle.
 - Opening hook and research motivation.
 - Condensed CRISP-DM narrative.
 - Selected readable charts with captions.
 - Simplified explanation of regression and ranking metrics.
 - Final model comparison.
 - Limitations and responsible-use section.
 - GitHub link placeholder.
 - YouTube link placeholder.
 - AI-assistance disclosure.
 - Final Medium publication URL.
 - Submission of the published article.
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3. YouTube walkthrough work

Verified YouTube outline or script: **none**.

No local file contains “YouTube,” “walkthrough,” or a video outline.

The official five-page assignment PDF places screencast/GitHub material under the later Assignment 2/extra-credit section. However, the expanded NBA project brief explicitly requests a YouTube walkthrough outline, so it remains part of the planned enhanced submission.

Missing:

- Walkthrough outline.
 - Demonstration sequence.
 - Suggested speaking notes.
 - Chart and notebook screen plan.
 - Recording.
 - Captions or transcript.
 - YouTube upload.
 - Public or unlisted URL.
 - README and Medium cross-links.
-

4. Chat-transcript export

Current NBA-project transcript export: **missing**.

The saved file is an instructor/example transcript about the Adult Income dataset. Its contents are not this NBA MVP project and it must not be submitted as the project transcript.

Missing:

- Export this NBA MVP conversation.
- Verify the export includes prompts, code, outputs, graphs, corrections, and decisions.
- Ensure no private information is visible.
- Save the final PDF with an unambiguous NBA-project filename.
- Submit it through the required course channel.

This is an explicit requirement in the official Assignment 1 PDF.

5. Actions you must complete manually

- Confirm the exact Kaggle dataset page and attribution.
 - Decide whether dataset redistribution is permitted; otherwise omit the ZIP.
 - Create or authorize the GitHub repository and push the final files.
 - Review the notebook and README in your own words.
 - Review and publish the Medium article from your Medium account.
 - Record and upload the YouTube walkthrough.
 - Export the complete NBA ChatGPT transcript near the end of the project.
 - Insert final GitHub, Medium, and YouTube URLs into all placeholders.
 - Verify repository visibility and link permissions.
 - Submit the required links/files to the LMS.
 - Confirm whether the instructor expects YouTube/GitHub for Assignment 1 or only as enhanced/extra-credit work.
 - Check the final submission for credentials, tokens, personal data, or unintended uploaded datasets.
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6. Part 2 replication work

Verified Part 2 replication artifacts: **none**.

Neither the official Assignment 1 PDF nor the expanded NBA project brief available here defines a “Part 2 replication” requirement. Searches for replication instructions did not locate a relevant brief.

Current status:

- Part 2 instructions are not available.
- No replication dataset is identified.
- No comparison target, student project, paper, or repository is identified.
- No replication notebook exists.
- No replication results or discrepancy analysis exists.
- No Part 2 report section exists.

Before this work can be assessed or started, you must provide or locate the exact Part 2 prompt.

Recommended completion order

1. Obtain the exact Part 2 instructions and Kaggle dataset URL.
2. Create the GitHub-ready Assignment-1/ structure, README, .gitignore, source attribution, and requirements.

3. Consolidate the 22 scripts into a reproducible notebook and stable supporting modules.
 4. Save the important analytical tables and repair the corrupted model artifact.
 5. Run the entire repository in a clean environment and reconcile notebook, README, report, and model results.
 6. Draft the Medium article, select its figures, and review it in your own voice.
 7. Create the YouTube outline, then record and upload the walkthrough.
 8. Push the final GitHub repository and add its URL to the article/video materials.
 9. Export the completed NBA-project chat transcript after the substantive work is finished.
 10. Publish Medium, finalize YouTube links, and submit all required materials.
 11. Complete Part 2 separately once its authoritative instructions are available.
- This audit stops here, with no presentation or file changes made.

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You

ok i have the repo ready with which to put all the programming and artifacts needed. but i would like you to generate the medium article and the readme for the repo. after that I will put whatever files are necessary in the repo, and you will help write a script for the youtube video presentations. after that video is recorded i will export this conversation. if there is anything else missing please let me know, if not go through with finalizing all the programing and what will be needed for the repo

ChatGPT

The GitHub-ready package is complete and verified.

Download the complete repository folder:

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Key deliverables:

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The package contains 72 files, including:

- 23 Python programs covering the incremental analysis and final workflow
- Executable orchestration through `run_all.py`
- Machine-readable result and ranking tables
- Selected publication-quality figures
- Deployment and schema-audit programs
- Verified 15.97 MB production model artifact
- README, Medium draft, model card, `.gitignore`, requirements, and data attribution
- No redistributed Kaggle dataset or credentials

Verification completed:

- Locked-test results reproduced exactly
- All six prespecified success criteria passed
- All Python files compiled successfully
- All six notebook code cells executed successfully against the uploaded data
- Complete reproduction command passed
- Model artifact deserialized successfully
- Target-leakage check passed
- Model SHA-256 matched the documented checksum
- Markdown links and ZIP integrity passed
- Credential-pattern scan passed

The verified data source is [1982–2022 NBA Player Statistics with MVP Votes](#).

Remaining manual work

1. Extract or copy Assignment-1/ into your repository.
2. Replace these placeholders: `<YOUR-GITHUB-REPOSITORY-URL>` `<ADD-GITHUB-URL>` `<ADD-PUBLISHED-MEDIUM-URL>` `<ADD-YOUTUBE-URL-AFTER-RECORDING>`
3. Review the Medium draft and adjust it into your personal voice before publishing.
4. Choose a repository license if your course expects one; that choice should be yours.
5. Push the repository and verify that its relative image links render on GitHub.
6. Publish the Medium article and add its final URL to the README.
7. Return for the YouTube walkthrough script.
8. Record and publish the video, then add its URL.
9. Export the conversation transcript as planned.
10. Provide the Part 2 instructions if Part 2 is required; no authoritative Part 2 replication specification has been supplied yet.

No presentation or YouTube script was created, consistent with your requested order.

Diagram options

The next stage is your repository upload and Medium review. After that, we can build the YouTube walkthrough script around the actual repository structure and final figures.